

Review: Seven Theorems Proving Optimal Search Efficiency for Two-Phase Search

Reviewer Comments — Pre-Publication Draft, February 3, 2026

Overall Impression

The core idea is sound and practically motivated — using exponential search to bracket the plateau, then binary search to refine, is a natural and elegant approach for monotonic functions. The theoretical framework is largely coherent. That said, there are some gaps that a sharp reviewer at NeurIPS or ICML will catch.

Strengths

Clear problem decomposition.

Breaking the analysis into seven focused theorems is pedagogically effective. Each theorem handles one concern (complexity, correctness, noise, optimality), which makes the argument easy to follow.

Theorem 7 is the crown jewel.

The information-theoretic lower bound is the most important result — it's what elevates this from "here's a fast algorithm" to "here's an optimal algorithm." The adversarial argument is sketched correctly.

Practical grounding.

The Covertypes example (99 evaluations $\rightarrow$ ~ 13 , $7.6\times$ speedup) makes the theoretical gains concrete, which reviewers appreciate.

Concerns and Weaknesses

1. The Discreteness Problem (Most Critical)

The search space is defined as integers $k \in [k_{\min}, k_{\max}]$, but the proofs largely treat $f(k)$ as a continuous function, invoking Lipschitz continuity and binary search on a continuous interval. This is a meaningful tension.

- Binary search on a discrete domain terminates in exactly $\lceil \log_2(N) \rceil$ steps — this should be stated cleanly rather than inheriting the continuous precision- δ framing.
- The Lipschitz constant L is well-defined for continuous functions, but for an integer-domain function you would typically need to argue it over the discrete setting. The connection needs a sentence or two of justification.

2. Theorem 2 and the Plateau Assumption

The plateau detection theorem assumes the performance curve becomes flat, but in practice Random Forest accuracy curves often have diminishing returns with low-level noise that never fully flattens. The assumption $f'(k) \approx 0$ for $k \geq k_p$ needs to be operationalized more carefully — what does " ≈ 0 " mean formally? The threshold ϵ in the windowed gradient criterion does this implicitly, but it should be stated explicitly as an assumption rather than buried.

3. Theorem 6 Has a Factor-of-2 Inconsistency

The bound in the statement is $\exp(-w\epsilon^2/2\sigma^2)$, but the derivation produces $2 \cdot \exp(-w\epsilon^2/2\sigma^2)$ before saying "for large w , simplifies to..." This is a sloppy step. A reviewer will flag it. Either carry the factor of 2 through or justify dropping it more rigorously (e.g., absorb into a constant for the asymptotic bound).

4. The Noise Model May Be Misspecified

The Gaussian noise assumption $\eta \sim N(0, \sigma^2)$ is stated in the setup, but cross-validation variance in Random Forests is not generally Gaussian — it's bounded (accuracy $\in [0,1]$) and often skewed, particularly for imbalanced datasets. Hoeffding's inequality is the right tool for bounded random variables, but it is being applied to a Gaussian model.

Critically, the setup states $|\eta| \leq \sigma$ (bounded noise), but Theorem 6 switches to $\eta \sim N(0, \sigma^2)$. This is an internal inconsistency that must be resolved before submission.

5. Theorem 7's "Comparison Model" Assumption

The lower bound proof assumes each evaluation yields exactly 1 bit. This is standard in comparison-based sorting lower bounds, but here you're evaluating a real-valued function — you get a real number, not just a binary comparison. A reviewer could argue you're underselling your information per query, which would weaken the lower bound argument. You should explicitly argue why, under adversarial construction, only 1 bit of useful information is extractable per query.

6. The Monotonicity Assumption Is Strong

The entire framework rests on monotonic-then-plateau behavior. This is stated as Assumption 1, which is fine, but the document would benefit from a short empirical justification or citation that Random Forest accuracy on benchmark datasets actually exhibits this property. Otherwise, the practical claims rest on an unvalidated assumption.

Minor Issues

- The summary table lists Theorem 2's bound as $e^{-(w\epsilon^2/8\sigma^2)}$ while Theorem 6 gives $e^{-(w\epsilon^2/2\sigma^2)}$. These are different — the document should clarify which theorem applies to which scenario, or unify them.

- "Phase 1 Exponential Search" should note that it doubles k , not that it performs generic exponential search — the specific doubling schedule $k_i = k_{\min} \cdot 2^i$ should be highlighted as a design choice.
- The conclusion mentions "Two-Phase Golden Number Search" but the document never defines or uses the "Golden Number" terminology anywhere. Either drop it or introduce it.

Verdict on Publication Readiness

The theoretical skeleton is solid enough to build on, but this is not ready for submission to NeurIPS or ICML in its current state. The internal inconsistency in the noise model (bounded vs. Gaussian) alone would likely trigger rejection.

Recommended fixes before resubmission:

1. Resolve the noise model inconsistency — stick with bounded noise, as it is more honest for CV variance.
2. Address the discrete vs. continuous domain tension explicitly.
3. Tighten the Theorem 6 proof to eliminate the factor-of-2 hand-wave.
4. Add empirical validation that the monotonicity assumption holds on your target datasets.

With those fixes, the theoretical contribution is genuine and the practical speedup is compelling. This could be a strong workshop paper or a solid contribution to a venue like AutoML or a hyperparameter optimization workshop as a stepping stone.

Review prepared March 20, 2026